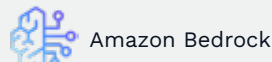
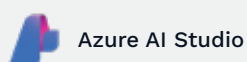
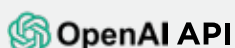
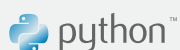




CERTIFICATE PROGRAM IN AI AND AGENTIC AI ENGINEERING

14 WEEKS ONLINE | LIVE MENTORSHIP SESSIONS BY INDUSTRY
EXPERTS | MONTHLY MASTERCLASSES BY JHU FACULTY

20+ Tools and Techniques Covered:



JOHNS HOPKINS UNIVERSITY ADVANTAGE

Johns Hopkins University (JHU) is a prestigious institution consistently ranked among the top 10 universities in the U.S. and has a strong reputation in research and education. JHU is well-positioned to offer compelling differentiators for its Continuing Education Programs.

WORLD-CLASS FACULTY EXPERTISE

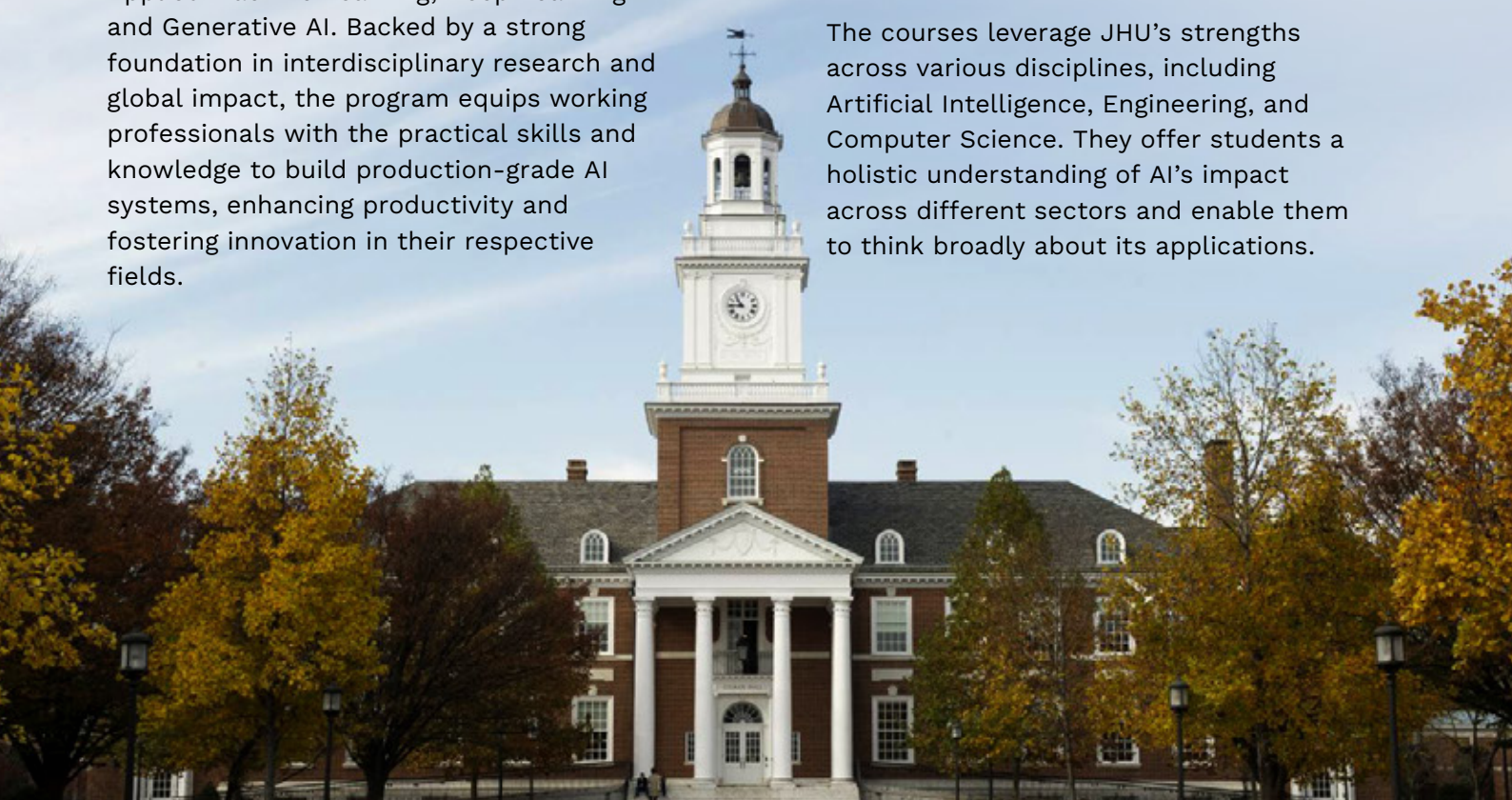
The courses are taught by faculty who are distinguished in academia and have industry experience leading Artificial Intelligence practices at Fortune 500 companies. This blend of academic rigor and industry expertise offers students unique insights into the practical applications of AI. Johns Hopkins University stands at the forefront of AI innovation, with faculty who are recognized leaders in Artificial Intelligence, Applied Machine Learning, Deep Learning and Generative AI. Backed by a strong foundation in interdisciplinary research and global impact, the program equips working professionals with the practical skills and knowledge to build production-grade AI systems, enhancing productivity and fostering innovation in their respective fields.

RESEARCH-DRIVEN CURRICULUM

The curriculum is designed based on the latest research and developments in AI, ensuring that students learn the most up-to-date and relevant information. Johns Hopkins University is the leading institution in research, consistently conducting nearly twice the R&D of any other university, and has maintained this unparalleled position for decades.

INTERDISCIPLINARY APPROACH

The courses leverage JHU's strengths across various disciplines, including Artificial Intelligence, Engineering, and Computer Science. They offer students a holistic understanding of AI's impact across different sectors and enable them to think broadly about its applications.





SCALE AI TO DRIVE INNOVATION

According to the LinkedIn Economic Graph Report 2026, AI Engineer is the fastest-growing job role in the U.S. Over three-quarters of IT roles require technical AI capabilities, highlighting the essential role AI engineers play in driving innovation and operational excellence across industries. Scaling AI solutions effectively is now key to meeting growing business demands and maintaining a competitive edge.

The 14-week **Certificate Program in AI and Agentic AI Engineering** from Johns Hopkins Whiting School of Engineering offers a critical advantage by equipping working professionals with the practical skills and knowledge to build production-grade AI systems. The comprehensive curriculum, designed and delivered by JHU faculty, covers essential topics such as data management, model training and deployment, MLOps, monitoring, and LLMOps for Generative AI and Agentic AI applications.

Through weekly live sessions with industry mentors, monthly masterclasses led by Johns Hopkins University faculty, hands-on projects, and real-world case studies, learners gain the expertise to operationalize AI workflows, manage AI lifecycle challenges, and deliver scalable, secure AI solutions. Designed for professionals in AI, data science, tech, or adjacent roles, this program prepares learners to confidently lead AI initiatives, accelerate career growth, and earn a globally recognized credential in the rapidly evolving AI engineering landscape.



#1 Fastest-Growing Job Category

AI Engineer is the fastest-growing job on LinkedIn

Source: LinkedIn Economic Graph Report 2026



83% of Companies List AI as a Top Priority

Highlighting AI's role in shaping their business plans

Source: Grand View Research



7 Out of 10 Fastest-Growing IT Roles Are Linked to AI

Across high-growth roles like AI and ML engineering positions

Source: AI Workforce Consortium Study, 2025

PROGRAM HIGHLIGHTS



Duration: 14 Weeks



Format: Delivered online, featuring recorded video lectures, weekly live sessions with global industry experts, and monthly masterclasses led by Johns Hopkins University faculty



Hands-on Learning: Build industry-ready skills to design, deploy, and operate AI systems in production through Great Learning cloud labs, hands-on projects, and real-world case studies



Learning Support: Get personalized assistance from a dedicated Program Manager and academic support through the Great Learning community, project discussion forums, and peer groups. Learners will also receive AI-assisted learning support through Great Learning

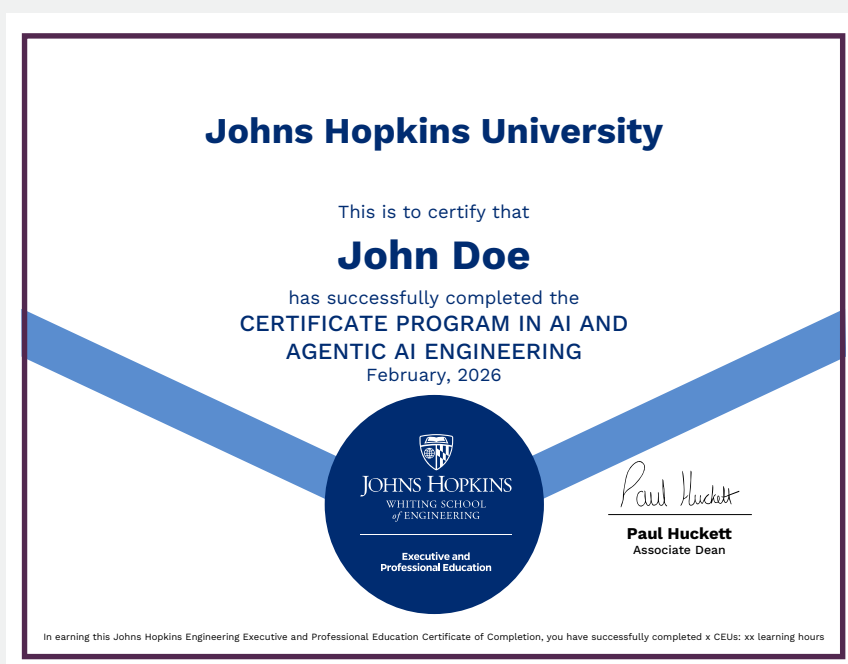


Create a Compelling ePortfolio: Build an industry-ready portfolio to showcase your mastery of skills and tools



Earn Recognized Credentials: Upon program completion, earn a globally recognized Certificate of Completion and Continuing Education Units (CEUs) from Johns Hopkins University

EARN A CERTIFICATE OF COMPLETION FROM JOHNS HOPKINS UNIVERSITY

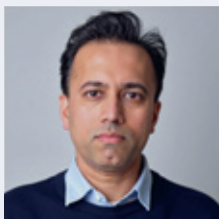


The image is for illustrative purposes only. The actual certificate may be subject to change at the discretion of Johns Hopkins University.



DELIVERED BY WORLD-CLASS FACULTY

The program is led by world-class Johns Hopkins University faculty with deep expertise at the intersection of Artificial Intelligence, AI strategy, Machine Learning, and applied analytics. The faculty member has designed and will deliver the curriculum while conducting masterclasses to ensure a comprehensive learning experience.



Kiran Chittargi

Adjunct Faculty - Computer Science, Johns Hopkins University

Kiran Chittargi is a Program Director at Microsoft and the Global Expansion Lead for Azure, where he spearheads strategies for launching and scaling Azure data centers worldwide, overseeing the entire lifecycle from opportunity assessment to deployment while collaborating with cross-functional teams. With over 20 years of experience in software engineering, he has deep expertise in designing and managing large-scale enterprise systems, working with technologies such as Big Data, Cloud Services (Azure and AWS), JavaScript, Java, and various web-based and open-source platforms. Kiran holds a Bachelor's in Industrial Engineering and a Master's in Information Systems. A passionate educator, he received the "Excellence in Teaching" award from Johns Hopkins University and has taught graduate courses since 2005. An active developer community member, Kiran frequently speaks at conferences and seminars, sharing his insights and expertise with professionals.

PROGRAM STRUCTURE

The Certificate Program in AI and Agentic AI Engineering is a 14-week online learning journey designed to equip learners with the skills to scale AI applications responsibly, accelerate career growth, and build credibility in the evolving AI landscape.

14-WEEK PROGRAM	
PRE-WORK	PREPARATORY MODULE
3 WEEKS	FOUNDATIONS OF AI OPERATIONS (AIOPS) Build a strong foundation in AIOps frameworks, operationalizing data, and model training and evaluation period
3 WEEKS	PRODUCTION-GRADE DEPLOYMENT WITH MLOPS Build production-ready MLOps skills to deploy, monitor, and manage AI models reliably at scale.
1 WEEK	LEARNING BREAK
1 WEEK	PROJECT WEEK
5 WEEKS	LLMOPS AND RESPONSIBLE AI Build skills to manage and scale Generative AI and Agentic AI solutions by applying LLMOps practices and responsible AI principles.
1 WEEK	PROJECT WEEK

PROGRAM CURRICULUM

The curriculum is designed by faculty at Johns Hopkins University. It is taught by a best-in-class professor and practicing AI professionals. The program covers data management, model training and deployment, MLOps, monitoring, and LLMOps for production-grade Generative and Agentic AI applications.

PRE-WORK

This module covers foundational cloud concepts, introduces the cloud ecosystem, guides learners in setting up cloud environments, and provides an overview of real-world AI applications on the cloud.

- Introduction to Cloud Concepts
- Understanding Cloud Ecosystem
- Real-World Use Cases for AI on Cloud



MODULE 1: AI OPS FOUNDATIONS

This module lays the groundwork for AI operations by covering the fundamentals of Artificial Intelligence and Machine Learning. It introduces key concepts, drivers, and frameworks for AIOps, with a deep dive into operationalizing data for AI and ML model training and evaluation cycle.

Week 01: Introduction to AI Ops and AI Engineering

Week 02: Operationalizing Data for AI

Week 03: Model Training and Evaluation

MODULE 2: PRODUCTION-GRADE DEPLOYMENT WITH MLOPS

This module focuses on the practical implementation of MLOps, covering the architecture and components of MLOps pipelines, including CI/CD principles and deployment strategies. Key topics include model monitoring, degradation, drift detection, and ensuring version control and reproducibility in production environments.

Week 04: MLOps Fundamentals and Model Deployment

Week 05: Building MLOps Pipelines

Week 06: Monitoring and Performance Analysis in AI Systems

Week 07: Learning Break

Week 08: Project 01

MODULE 03: LLMOPS AND RESPONSIBLE AI USAGE

This module focuses on operationalizing and responsibly using Generative and Agentic AI. It introduces LLMops fundamentals, including CI/CD implementation and monitoring for LLMs, as well as the deployment of Agentic AI systems. The module also covers critical aspects of evaluating, securing, and ensuring responsible use of these AI systems.

Week 09: LLMops Fundamentals

Week 10: LLMops in Production

Week 11: Agentic AI in Production

Week 12: Evaluating and Securing Agentic AI Systems

Week 13: Responsible AI and AIOps Adoption

Week 14: Project 02

SAMPLE CASE STUDIES

IT OPERATIONS

Title: Proactive Incident Management with AIOps

Objective: Leverage AIOps practices to transform IT operations, enabling proactive incident management, reduced downtime, and scalable, reliable service delivery.

Skills: AIOps implementation, Incident management, IT operations automation, Service reliability and scalability

RETAIL

Title: Predicting Purchase Propensity from Retail Transaction Data

Objective: Train and evaluate an ML model to predict customer purchase likelihood from historical transaction data, enabling targeted marketing, improved conversion rates, and optimized campaign ROI.

Skills: Predictive Modeling, Customer Segmentation, Machine Learning Evaluation, Marketing Analytics

HEALTHCARE

Title: Hospital Re-admission Risk Prediction System

Objective: Develop an end-to-end MLOps pipeline for hospital re-admission risk prediction, incorporating experiment tracking, version control, and CI/CD automation to ensure a reproducible, reliable, and continuously improving clinical decision support system.

Skills: MLOps Pipeline Development, Experiment Tracking, CI/CD Automation, Clinical Decision Support

ENERGY

Title: Energy Demand Forecasting System

Objective: Develop an MLOps-driven energy demand forecasting system with performance monitoring and drift detection to ensure accurate predictions under changing consumption patterns and seasonal shifts.

Skills: MLOps Pipeline Development, Performance Monitoring, Drift Detection, Forecasting Analytics



SAMPLE PROJECTS

FINANCIAL SERVICES

Title: Credit Default Risk Scoring

Objective: Build an automated MLOps pipeline that predicts borrower default risk from financial and behavioral data, enabling data-driven lending decisions through a scalable, production-ready ML system.

Skills: Python, Pandas, Scikit-Learn, MLflow, Random Forest, XGBoost, Model Deployment, MLOps, CI/CD, Experimentation Tracking, Business Recommendations

RETAIL

Title: Agentic AI-Powered Retail Shopping Assistant

Objective: Develop a shopping assistant powered by Agentic AI that understands customer intent, retrieves relevant data, executes actions like order tracking, and delivers personalized support through a production-ready LLMOps system.

Skills: Python, Large Language Models, Prompt Engineering, LangChain, OpenAI, LangGraph, Agentic AI

WHO IS THIS PROGRAM FOR?



AI/ML/DL Practitioners: Professionals who have built models or POCs in Python and want to learn how to operationalize them at scale using MLOps, LLMOps, and AIOps best practices.



GenAI and Agentic AI Builders: Individuals who have experimented with LLMs, OpenAI APIs, or agentic workflows and want to advance to secure, monitored, and reliable production deployments.



Data and Analytics Professionals: Practitioners familiar with modeling and data pipelines who want to learn about the end-to-end AI lifecycle, including deployment, monitoring, and automation.



Technology Professionals, Architects and Leaders: Leaders overseeing AI initiatives who want a structured framework to develop and scale AI and Agentic AI systems in real-world environments.

LEARNING OUTCOMES



Design and implement AI/ML systems on the cloud by building end-to-end MLOps pipelines.



Operationalize and scale Generative AI and Agentic AI solutions in production by applying LLMOps principles to design, deploy, and manage LLM-powered workflows on cloud platforms.



Evaluate, monitor, and secure Agentic AI systems by defining and tracking performance metrics for continuous evaluation, implementing observability and security measures.



Apply responsible AI practices and data security controls by integrating governance, compliance, privacy, and security principles into the design, deployment, and operation of AI solutions.

PROGRAM PEDAGOGY

- ✓ **Program Delivery:** This 14-week, fully online program is delivered through a combination of live mentorship sessions with industry experts, masterclasses with best-in-class JHU faculty, hands-on projects and case studies on real-world scenarios, quizzes, and self-paced modules.
- ✓ **Live Mentorship and Masterclasses:** The program includes live mentorship by industry experts and live faculty-led masterclasses through a unique learning model that keeps learners on track with career goals.
- ✓ **AI-Assisted Learning:** AI-powered tools support video content, project work, and practice activities to enhance learning efficiency and engagement.
- ✓ **Hands-on Learning:** This program helps learners drive business value by effectively using AI technologies through industry-relevant tools and technologies, hands-on projects, and real-world case studies from different industry verticals to reinforce conceptual and practical understanding.
- ✓ **Industry-Aligned Curriculum:** The curriculum covers key areas including data management, model training and deployment, MLOps, monitoring, and LLMOps for production-grade Generative AI and Agentic AI applications.
- ✓ **Dedicated Program Manager:** A Program Manager assists learners through their learning journey to ensure they achieve their learning objectives. They will act as the sole point of contact during the program, supporting learners by ensuring timely assistance from the ecosystem. Along with monitoring progress, they offer necessary encouragement to ensure success.



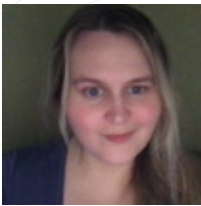
MENTORSHIP

Stay on track with your learning goals through personalized mentorship from industry experts. Engage in curated micro-classes and interactive live sessions designed to deepen your understanding, clarify doubts, and apply concepts through hands-on, real-world projects. This structured support ensures your learning remains relevant, practical, and aligned with industry needs.



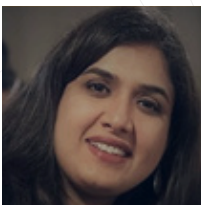
HASSAN AYMAN

Senior Data Scientist,
IBM



TINA KOVACOVA

Data Advisor,
Kaggle



AROOJ AHMED QURESHI

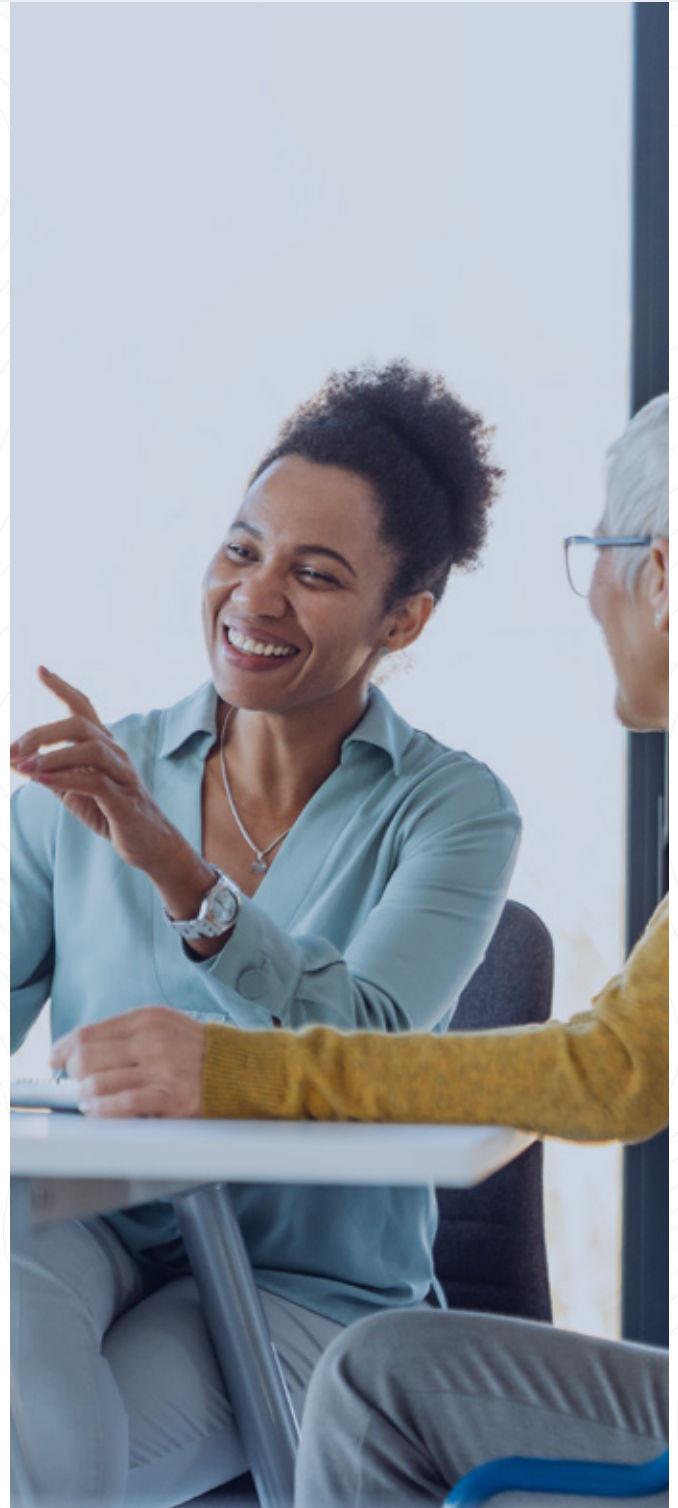
Senior Manager AI Engineering,
Bell



ISHWOR BHUSAL

Data Scientist - Supply Chain
Data Innovation,
Nissan Motor Corporation

Note: This is an indicative list and is subject to change based on the availability of mentors.



CAREER SUPPORT

The program offers career guidance designed to help translate program learning into professional advancement.



Resume Builder: Access to an AI-powered resume builder to create professional, impactful resumes aligned with targeted roles.



Mock Interviews: Unlimited mock interviews powered by AI to practice and refine interview skills.

ADMISSION PROCESS

The admission process is conducted on a rolling basis and will be closed once the requisite number of candidates is enrolled.



APPLY

Fill out an online application form



REVIEW

Eligible applications will be reviewed by a panel from Great Learning



JOIN PROGRAM

An offer letter will be sent to the selected candidates

PROGRAM FEE

\$2100

Please contact the Program Advisor from Great Learning for more information on offers, payment plans, and eligibility for financial assistance.

PROGRAM PARTNER



Great Learning is a leading global ed-tech company for professional and higher education. It offers comprehensive, industry-relevant programs across various cutting-edge Technology, Data, and Business domains. These programs are developed in collaboration with the world's foremost academic institutions, such as Johns Hopkins University, MIT Professional Education, The University of Texas at Austin, Northwestern School of Professional Studies, Deakin University, Great Lakes Institute of Management, and more. They are constantly reimagined and revamped to address the dynamic needs of the industry. Offered in blended, classroom, and purely online modes, these programs are delivered with the help of expert mentors and highly qualified faculty. Great Learning is on a mission to enable transformative learning and career success in the digital economy, which has impacted 14 million+ learners from over 170 countries.



14 Mn+
Learners



170+
Countries



8100+
Industry Experts



SCALE SMARTER WITH AI

CONTACT US

ai-engineering.jhu@mygreatlearning.com
+1 410 680 3011
